

Technologies Used – Malaeb Hail Platform

This section summarizes the main technologies and tools used to develop the Malaeb Hail platform.

1) Web Development

- Python – main backend programming language.
- Flask – lightweight web framework for routes and server logic.
- HTML5 – page structure.
- Jinja2 – Flask templating engine.
- JavaScript – booking interaction, map focus, and Tawakkalna login simulation.

2) Database

- SQLite – lightweight file-based database.
- Database file: malaeb.db
- Main tables: parks and bookings.
- Python sqlite3 module for data access.

3) UI Design

- CSS3 – styling, colors, and responsive layout.
- RTL Arabic interface design.
- Cairo / Arabic-friendly typography.
- Color palette inspired by Hail Municipality branding (greens).
- Flexbox and Grid for booking and dashboard layouts.
- Google Maps Embed (iframe) for park location and pin.

4) Version Control

- Git – source code version tracking.
- Optional remote hosting on GitHub/GitLab for submission and backup.

5) Local Server

- Flask Development Server – local runtime during development.
- Local URL: `http://127.0.0.1:5000`
- Debug mode for testing and auto-reload.

Summary Table

Category	Technology / Tool	Purpose
Web Development	Python + Flask	Backend server and routes
Web Development	HTML + Jinja2 + JavaScript	Pages and interactivity
Database	SQLite (malaeb.db)	Store parks and bookings
UI Design	CSS3 + RTL + Cairo	Arabic modern interface
UI Design	Google Maps iframe	Map and location pin
Version Control	Git	Track project versions
Local Server	Flask Local Server	Run the app locally